

JustHumanResource Technical Documentation

1. Executive Summary

JustHumanResource is a comprehensive Human Resource Management System (HRMS) built using modern Java technologies. The system provides a robust platform for managing employee lifecycles, payroll processing, performance appraisals (KPIs), and various HR workflows. It leverages enterprise-grade tools like Flowable for process orchestration and Keycloak for secure identity and access management.

2. Technology Stack

- **Backend:** Java 21, Spring Boot 3.5.3
- **Database:** PostgreSQL
- **Workflow Engine:** Flowable 7.1.0
- **Security:** Keycloak (OAuth2/OpenID Connect)
- **Frontend:** Thymeleaf, HTMX, Bootstrap 5.3
- **API Documentation:** SpringDoc OpenAPI (Swagger)
- **AI Integration:** Spring AI (OpenAI)
- **Payment Gateway:** Paystack
- **Build & Deployment:** Maven, Jib (Containerization)

3. System Architecture

The application follows a modular monolith architecture. Each business domain is encapsulated within its own package, promoting separation of concerns and maintainability.

Core Architecture Components:

- **Controllers:** Handle HTTP requests and interact with services.
- **Services:** Implement business logic and orchestrate data flow between repositories and other components.
- **Repositories:** Spring Data JPA interfaces for database interaction.
- **Entities:** JPA models representing the database schema.
- **DTOs & Mappers:** Data Transfer Objects for API/Web layer, with MapStruct for efficient mapping.
- **Workflows:** BPMN 2.0 processes managed by Flowable for complex multi-step HR operations.

4. Module Overview

4.1. Human Resources (HR)

Managed under the `com.justjava.humanresource.hr` package.

- **Employee Management:** Handles personal details, bank accounts, documents, and position history.
- **Organization Structure:** Manages Departments, Job Grades, Job Steps, and Pay Groups.
- **Onboarding:** Orchestrates the new hire process through a dedicated Flowable workflow.

4.2. Payroll Management

Managed under the `com.justjava.humanresource.payroll` package.

- **Components:** Defines Allowances, Deductions, and Tax Reliefs.
- **Processing:** Supports payroll cycles, period closing, and automated payroll runs.
- **Reports:** Generates Pay Slips, Payroll Audit Logs, and Reconciliation Reports.
- **Integration:** Uses Paystack for payment processing.

4.3. Workflow Management

Managed under the `com.justjava.humanresource.workflow` package.

- **Flowable Integration:** A custom `FlowableTaskService` provides a simplified API for interacting with the Flowable engine.
- **Key Processes:**
 - `onboardingProcess` : Manages new employee entry.
 - `payrollApproval` : Multi-stage approval for payroll runs.
 - `employeeAppraisalProcess` : Performance review cycle.
 - `payrollPeriodCloseProcess` : End-of-month processing.

4.4. Performance Management (KPI)

Managed under the `com.justjava.humanresource.kpi` package.

- Supports monthly and quarterly appraisal cycles.
- Orchestrates reassignment and evaluation processes via BPMN workflows.

4.5. Security & Authentication

- **Keycloak:** Used as the Identity Provider (IdP).
- **OAuth2:** Secure authentication for both Web and Mobile clients.
- **Multi-Realm Support:** Configured for different authentication realms (e.g., `humanResources`, `mobile-auth-realm`).

5. Key Workflows (BPMN)

The project includes several critical business processes defined in `src/main/resources/processes` :

- **Batch Payroll Process:** Handles large-scale payroll calculations.
- **Onboarding Process:** Steps for integrating new employees.
- **KPI Evaluation:** Automated performance review stages.
- **Payroll Approval:** Ensures compliance and oversight before disbursement.

6. Setup and Deployment

6.1. Prerequisites

- JDK 21
- PostgreSQL
- Keycloak Server
- Maven

6.2. Configuration

Configuration is managed via `application.yml` and environment variables. Key variables include:

- `JDBC_DATABASE_URL`
- `ISSUER_URI` (Keycloak)
- `openaiAPI_KEY`
- `KEYCLOAK_CLIENT_SECRET`

6.3. Building the Application

```
./mvnw clean package
```

6.4. Containerization

The project uses Jib to build OCI-compliant images without a Docker daemon:

```
./mvnw jib:build
```

7. API Endpoints

Comprehensive API documentation is available at `/swagger-ui.html` when the application is running, covering all modules including HR, Payroll, and Workflows.